



Challenges of Integrating AI into the Internal Audit Functions of the Public Universities in Ghana



Jonathan Casely Wiredu^{*}, Mishelle Doorasamy, Kiran Baldavoo

School of Accounting, Economics and Finance, College of Law and Management Studies, University of KwaZulu-Natal, 4000 Durban, South Africa

*Correspondence: Jonathan Casely Wiredu (224193361@stu.ukzn.ac.za)

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Abstract: This study examined the challenges faced by internal auditors in adopting AI for the internal audit functions of the public universities in Ghana. The study used a qualitative research design that involved semi-structured interviews with six audit professionals from six prominent public universities; it was guided by the Technology-Organization-Environment (TOE) and Diffusion of Innovation (DOI) theory. The study employed NVivo 14 software to analyse data thematically. The findings revealed four critical themes influencing AI adoption: technological readiness, organizational culture, capacity and competency gaps, and regulatory and ethical ambiguities. The most significant obstacles were identified as technological constraints, such as outmoded infrastructure and inadequate data systems. Furthermore, innovation was impeded by bureaucratic leadership structures and inadequate management commitment. The adoption of AI was further restricted by the ambiguities surrounding its ethical and regulatory use, as well as skill deficiencies. The study underscored the need for leadership commitment and governance innovation to realize the full potential of AI in public audit transformation. It contributes to the literature by contextualizing the challenges of AI adoption in the higher education sector of a developing economy, specifically Ghana, to offer theoretical insights into the intersection of digital readiness and institutional culture. For policymakers, it also provides practical recommendations such as targeted capacity building, infrastructure enhancement, and policy reforms to support AI-driven auditing.

Keywords: AI; Internal audit; Public universities; Technology-Organization-Environment framework; Diffusion of Innovation theory; Digital transformation

1. Introduction

The rapid advancement of technology has fundamentally transformed organizational processes across industries, and auditing is no exception. Over the past decade, internal audit functions have evolved from primarily compliance-focused activities to more dynamic, risk-based, and strategic contributions to organizational governance (Anomah et al., 2024; Dawuda & Salman, 2024). The integration of digital technologies, particularly AI, is a critical component of this evolution. AI has the potential to improve analytical capabilities, provide predictive insights, automate routine audit procedures, and enhance fraud detection (Oyedotun et al., 2025). Auditors can identify anomalies, forecast risks, and generate actionable insights by utilizing AI-enabled audit tools, which are more efficient than traditional methods in processing large volumes of structured and unstructured data (Aikins et al., 2022; Issah & Baah, 2025). The incorporation of AI in auditing necessitates auditors to acquire new competencies and adjust to the swiftly evolving technological landscapes, despite its potential. Current literature underscored that, despite the global transformation of auditing by AI, there is a substantial disparity in the comprehension of how auditors in developing economies, particularly in resource-constrained environments, interact with these technologies (Anomah et al., 2024; Tetteh et al., 2023). The scarcity of empirical evidence in these contexts raises concerns regarding the organizational factors that may impede the effective utilization of AI by auditors. This knowledge deficit is particularly severe in public sector institutions, where the accountability stakes are high, despite the significant variation in technological infrastructure, policy frameworks, and digital

literacy (Dawuda & Salman, 2024; Kudo, 2024). Addressing these gaps is critical for both theory and practice, as it can inform strategies to enhance audit quality, organizational governance, and sustainable integration of AI technologies.

In the public sector, and particularly within higher education institutions, internal auditing is essential for the prudent utilization of limited resources, regulatory compliance, and financial integrity (Abudu & Syed, 2025; Issah et al., 2024). Public universities in developing countries, such as Ghana, encounter distinctive operational challenges, such as budgetary constraints, bureaucratic red tape, inadequate staffing, and more intense scrutiny from a variety of stakeholders, including government bodies, students, and external auditors (Afadzinu & Koltai, 2025). The incorporation of AI into the internal auditing of these universities has the potential to address some of these challenges by improving audit efficiency, facilitating proactive risk identification, and offering more profound insights into intricate financial and operational data (Issah et al., 2024). The literature suggested that the adoption of AI in the auditing of higher education was impeded by several gaps and obstacles, despite the advantages. These include a lack of formal policies for digital transformation, insufficient investment in technological infrastructure, limited technical expertise among the staff responsible for internal audit, and resistance to changes due to fear of job displacement or perceived complexity of AI systems (Adorm-Takyi, 2023; Celestin & Gidisu, 2024). Moreover, the challenges of AI integration are further complicated by regulatory ambiguities and ethical concerns regarding data privacy, AI-driven decision-making, and accountability (Abudu & Syed, 2025; Dawuda & Salman, 2024). These gaps highlight the need for empirical research to explore the real-world experiences of internal auditors in public universities, particularly in developing economies, where the potential benefits of AI may be constrained by structural, organizational, and human factors.

Despite the growing recognition of AI as a transformative force in auditing, its adoption and implementation within public universities in Africa remain strikingly underexplored. Existing scholarship of AI in auditing has predominantly focused on private sector organizations and developed economies, where technological infrastructure, digital literacy, and innovation cultures are comparatively advanced (Awuah et al., 2022; Smith et al., 2025). Consequently, limited attention has been paid to the distinct challenges faced by public sector institutions in developing contexts, such as Ghana, where bureaucratic structures, funding constraints, and weak technological ecosystems complicate digital transformation efforts (Afadzinu et al., 2024; Obuobi, 2024). Empirical studies examining how internal auditors in public universities navigated these challenges are particularly scarce and created a notable research gap. Furthermore, while the potential of AI to enhance fraud detection, risk management, and operational efficiency is widely acknowledged (Agana et al., 2025; Ataribanam, 2024), little is known about how internal auditors in the public education sector perceive, adopt, or operationalize these technologies. The absence of context-specific evidence obscures the understanding of how factors such as leadership support, institutional culture, data governance, and audit autonomy influence the outcomes of its adoption (Anokye, 2022; Yanuarisa et al., 2025). In addition, ethical and regulatory gaps found in data security, algorithmic transparency, accountability, and alignment with national audit standards remain largely unaddressed (Awuah et al., 2022; Ceki & Moloi, 2025). These shortcomings highlight a critical empirical gap in understanding how systemic, infrastructural, and human resource limitations constrain the integration of AI into audit practices in African public universities. Accordingly, this study sought to investigate the central question:

RQ: What challenges do internal auditors face in integrating AI into internal audit functions of the public universities in Ghana?

The paper proceeds as follows: Section 2 reviews the theoretical background and empirical studies. Section 3 explains the methodology; Section 4 presents and analyses results; Section 5 discusses findings and contributions, and Section 6 concludes with policy implications and suggestions for future research.

2. Theoretical Background

2.1 Technology-Organization-Environment Framework

The Technology-Organization-Environment (TOE) framework provided a robust and integrative perspective for understanding how organizations adopt and implement technological innovations. The framework asserted that the adoption and diffusion of technology were shaped by three interdependent contexts: technological, organizational, and environmental (Al-Okaily et al., 2024). The technological context is related to the perceived benefits, complexity, and compatibility of innovations, such as AI, with existing systems and workflows. In auditing, this refers to the extent to which AI tools enhance risk detection, efficiency, and data accuracy relative to traditional methods (Tetteh et al., 2021). The organizational context encompasses the structural and managerial capacity of an entity, including leadership commitment, financial resources, audit independence, and cultural openness to innovation (Al-Okaily et al., 2024). The environmental context involves the external factors that either facilitate or constrain adoption, such as regulatory compliance requirements, societal expectations for accountability, and competitive or policy pressures (Sarpong, 2021). The TOE framework assumed that organizations rationally evaluated costs, risks, and benefits before embracing new technologies, implying a linear

and objective decision-making process (Anim et al., 2025). However, this assumption may not be held in developing countries, where sociocultural values, informal power structures, and political influences often shape the decisions of adoption beyond rational evaluation (Doe-Dartey & Valand, 2024).

Critics of the TOE framework argued that its technocentric and organizational focus tended to downplay the human and cultural dimensions of innovation adoption, especially in public sector institutions where resistance to change and bureaucratic inertia are pervasive (Celestin & Gidisu, 2023b). In addition, the framework has been criticized for its assumption of homogeneity across organizations, which ignores the differences in institutional mandates and national contexts (Asmah & Kyobe, 2025). This study expanded the TOE framework by incorporating qualitative insights from internal auditors in the public universities in Ghana, a setting that is characterized by limited resources, expectations of high accountability, and evolving regulatory environments, in response to these limitations (Yanuarisa et al., 2025). This contextualized application demonstrated how technological and environmental factors, as well as cultural norms, leadership attitudes, and capacity constraints, interacted to influence the obstacles associated with the adoption of AI. The analysis of three critical dimensions in this study was informed by the TOE framework: technological readiness, organizational capacity, and environmental pressures. This multivariate approach offered a thorough comprehension of the confluence of contextual and systemic factors that either facilitate or impede the integration of AI into the internal audit functions of the public universities in Ghana.

2.2 Diffusion of Innovation Theory

The Diffusion of Innovation (DOI) theory offered an explanatory model for understanding how new ideas, technologies, or practices spread within a social system. The theory posited that the adoption of innovation depended on five perceived attributes: relative advantage, compatibility, complexity, trialability, and observability (Sheta et al., 2025). Relative advantage concerns the degree to which AI improves the outcomes of auditing compared to traditional methods, while compatibility measures its alignment with existing organizational values and practices. Complexity reflects the perceived difficulty in understanding or using AI tools, and trialability denotes opportunities for experimentation before full-scale implementation. Observability refers to the visibility of tangible benefits derived from adoption (Shuwaili et al., 2024). DOI assumed that the adoption of innovation followed a predictable sequence from knowledge and persuasion to decision, implementation, and confirmation, driven by individual perceptions and social influence (Sheta et al., 2025). In the auditing profession, DOI helps explain how auditors' cognitive orientations, professional skepticism, and ethical concerns influence their readiness to engage with AI-driven auditing tools (Anomah et al., 2024). However, this assumption may not reflect the realities of public institutions in developing contexts, where the decisions of adoption are often collective, bureaucratic, and influenced by hierarchical structures and regulatory directives rather than individual discretion (Dawuda & Salman, 2024).

The primary limitation of DOI is its inadequate consideration of institutional, environmental, and infrastructural constraints that frequently influence the adoption of technology in low-resource settings, although it offers valuable insights into human and behavioral factors that influence adoption (Oyedotun et al., 2025). It also has a propensity to view innovation as a uniform process, disregarding contextual variations in governance systems, leadership orientation, and resource availability (Issah & Baah, 2025). This study addressed these limitations by situating the diffusion of AI in the auditing context of Ghana's public universities, where individual acceptance interacts with institutional readiness and external policy mandates. By exploring auditors' attitudes, ethical considerations, and perceptions of risk and utility, this study captured the socio-behavioral dimensions of AI integration, which are often absent from purely structural models. DOI thus informs the examination of perceptual and behavioral variables such as auditor awareness, professional competence, perceived usefulness, and resistance to technological change. By contextualizing DOI within a public university environment, the study revealed that the innovative behavior of internal auditors was not merely an outcome of perceived advantage but also a reflection of institutional support, ethical constraints, and cultural attitudes toward technology. Consequently, DOI enriched this research by deepening the understanding of the human and social mechanisms that either facilitate or impede the adoption of AI in the internal audit functions of Ghanaian public universities.

2.3 The Rise of AI in the Evolution of Auditing

The auditing profession has undergone significant evolution over the past century, transitioning from traditional and manual verification of financial statements to the adoption of advanced and technology-driven approaches that emphasize risk management, predictive analysis, and strategic decision support. Initially, auditors relied on manual sampling techniques and paper-based records, which limited both efficiency and accuracy (Aikins et al., 2022). The scope of auditing was broadened to encompass automated data extraction, transaction testing, and continuous monitoring with the introduction of computer-assisted audit tools and data analytics in the late twentieth century (Tetteh et al., 2023; Kudo, 2024). The auditing landscape has been revolutionized by the recent emergence of AI

and machine learning, which have allowed systems to process extensive datasets, identify anomalies, and offer real-time assurance (Issah et al., 2024). AI-driven auditing tools are being used more frequently in areas such as fraud detection, compliance automation, predictive risk assessment, and performance analytics. This enables auditors to improve judgment-based tasks and reduce procedural inefficiencies (Afadzinu & Koltai, 2025). Empirical evidence suggested that the incorporation of AI improved anomaly detection, enhanced audit quality, and strengthened governance by providing data-driven insights (Abudu & Syed, 2025). Nevertheless, developing countries encounter obstacles such as infrastructure, technical competence, and institutional preparedness, although developed economies have made significant strides in integrating AI into audit practices (Aikins et al., 2022). Consequently, the development of auditing is indicative of not only technological advancement but also the increasing disparities in the adoption of innovation. This underscores the necessity of contextualized strategies to effectively integrate AI across a variety of economic environments.

2.4 AI in the Public Sector and Higher Education

In the public sector, AI has emerged as a transformative instrument to provide new opportunities for the improvement of governance, accountability, and service delivery by facilitating operational transparency and data-driven decision-making (Abudu & Syed, 2025). The applications of AI have been deployed more frequently in higher education to boost administrative efficiency, facilitate financial management, and guarantee quality assurance in academic institutions (Adorm-Takyi, 2023). AI has the potential to improve the efficiency of resource utilization and institutional accountability by streamlining auditing, budget allocation, and performance monitoring processes in public universities (Celestin & Gidisu, 2023a). Nevertheless, there is a dearth of research on the integration of AI in higher education, particularly in developing economies. In these regions, universities encounter bureaucratic rigidity, inadequate funding, and outmoded technological infrastructures that impede digital transformation (Tetteh et al., 2023). In contrast to private institutions, which frequently have access to advanced technologies and more flexibility, public universities are subject to intricate regulatory environments that impede innovation (Awuah et al., 2022). Moreover, the incorporation of AI into the administration of public education is impeded by challenges arising from its implementation, including a lack of technical expertise, insufficient digital literacy, and resistance to organizational change (Smith et al., 2025). In Africa, these obstacles are exacerbated by inadequate information and communication technology (ICT) infrastructure, inconsistent policy support, and insufficient institutional preparedness (Kudo, 2024). In Ghanaian public universities, the context-sensitive approaches to AI-driven governance and audit modernization in the public education sector are underscored by the unique challenges of adoption that are created by the intersection of fiscal constraints, centralized decision-making, and policy ambiguities (Afadzinu et al., 2024).

2.5 Integration of AI into Internal Auditing

The integration of AI into the internal audit function represents a paradigm shift from traditional compliance-based auditing to a more proactive, data-driven, and strategic approach that enhances governance and organizational resilience. From a conceptual standpoint, the incorporation of AI into auditing entails the utilization of intelligent algorithms, data analytics, and automation to facilitate audit planning, evidence collection, risk assessment, and reporting (Ataribanam, 2024). Auditors are becoming more competent in identifying emergent risks and irregularities in real time by utilizing AI tools in both corporate and governmental contexts to facilitate predictive analytics, continuous auditing, and anomaly detection (Ceki & Moloi, 2025). Empirical studies have shown that AI improved fraud prevention, reinforced compliance monitoring, and enhanced audit efficiency by processing immense datasets with greater speed and precision than manual methods (Obuobi, 2024). In addition, AI-driven auditing facilitates strategic decision-making by offering insights into operational trends and risk exposures (Agana et al., 2025). Despite these advantages, the successful integration of AI is contingent upon several critical factors, such as policy support frameworks, robust infrastructure, staff training, and leadership commitment (Afadzinu et al., 2024). Resistance to technological change and a lack of skills frequently impede effective implementation; therefore, the perceptions and competencies of internal auditors are critical to the outcomes of adoption (Ataribanam, 2024). In order to fully leverage the potential of AI to improve the internal audit function, it is imperative to cultivate digital readiness, ethical awareness, and institutional support, particularly in the public sector and universities.

2.6 Empirical Evidence from Developing Economies

Empirical studies from both developed and developing countries revealed diverse patterns in the adoption and impact of AI within auditing, accounting, and governance functions. In developed economies such as the United States, the United Kingdom, and parts of Europe, AI has become a transformative component of modern auditing, thus improving accuracy, efficiency, and predictive insight (Afadzinu et al., 2024; Obuobi, 2024). In these contexts,

research indicated that prominent auditing firms have implemented AI tools, including natural language processing, data mining, and machine learning algorithms, to improve fraud detection, automate risk assessments, and fortify compliance monitoring (Asmah & Kyobe, 2025). For example, Celestin & Gidisu (2023b) illustrated that AI-based auditing systems facilitated real-time data analysis, which enhanced decision-making and reduced the necessity of manual sampling. Similarly, research conducted by Anokye (2022) and Ceki & Moloi (2025) suggested that AI-assisted auditors in the identification of irregularities and potential misstatements with greater precision, resulting in an improvement in the quality of assurance. Al-Okaily et al. (2024) reported that the implementation of AI-driven continuous auditing platforms in Europe led to heightened transparency, strengthened corporate governance, and increased stakeholder confidence. These studies collectively underscored the fact that the integration of AI was significantly influenced by the availability of digital infrastructure, professional expertise, and supportive regulatory frameworks in technologically advanced economies.

In contrast, empirical evidence from developing economies depicted a slower and more complex adoption trend, largely constrained by infrastructural, financial, and institutional challenges. Studies across Sub-Saharan Africa, Asia, and Latin America suggested that while auditors recognized the potential of AI to revolutionize their work, the adoption rates remained low due to limitations of resources, weak institutional support, and insufficient technical capacity (Asmah & Kyobe, 2025). Doe-Dartey & Valand (2024) discovered that the majority of internal auditors in Nigeria had inadequate access to AI tools and lacked the necessary training to effectively employ them, leading to their continued dependence on manual audit techniques. In Kenya and South Africa, research demonstrated that the implementation of digital audit initiatives was impeded by fragmented ICT infrastructure and inadequate regulatory alignment, despite the existence of such initiatives (Sheta et al., 2025). Tetteh et al. (2021) noted that public institutions in Ghana frequently encountered competing budgetary priorities that restricted investment in emergent technologies, despite the increasing awareness of AI in auditing. Anim et al. (2025) and Sarpong (2021) also observed that bureaucratic governance structures and antiquated systems in public universities decreased the agility necessary for digital transformation. Empirical evidence from Asian economies, including India and Malaysia, is consistent with these trends. The adoption of AI in auditing was influenced by the perceived utility of technology, leadership support, and organizational capability (Doe-Dartey & Valand, 2024). Some developing nations are experiencing gradual progress despite these limitations. For instance, research conducted in the United Arab Emirates and Singapore indicated that the early implementation of AI in government auditing had enhanced financial supervision, compliance accuracy, and audit timeliness (Adorm-Takyi, 2023).

Although research on AI in auditing has grown in recent years, the majority of studies have focused on private firms and technologically advanced economies, resulting in a geographical and sectoral imbalance within the evidence base (Awuah et al., 2022; Celestin & Gidisu, 2023a). Research on public institutions in developing countries remains limited, despite ongoing infrastructural deficiencies, bureaucratic challenges, and underdeveloped innovation ecosystems that are likely to influence the adoption of AI in distinct ways (Afadzimu & Koltai, 2025; Tetteh et al., 2021). Existing research tended to emphasize technological factors, while neglecting the influence of organizational culture, leadership practices, and institutional governance on digital transformation in resource-limited settings (Anim et al., 2025; Doe-Dartey & Valand, 2024). Theoretical applications of the TOE and DOI frameworks in the auditing of the public sector remain underexplored, with limited research investigating how perceived complexity, regulatory ambiguity, security issues, and compatibility challenges converge in higher education environments (Issah & Baah, 2025; Shuwaili et al., 2024). Furthermore, personal experiences of internal auditors managing these constraints are significantly underrepresented in existing literature, leading to a limited understanding of context-specific obstacles to technological innovation. This research addressed these empirical, theoretical, and contextual voids by offering qualitative evidence from the public universities in Ghana regarding the multifaceted factors that hinder the integration of AI into internal audit functions.

3. Research Methodology

This study adopted a qualitative exploratory research design within the interpretivist paradigm to examine the challenges internal auditors face in integrating AI into the audit functions of public universities in Ghana. Given the limited empirical understanding of AI adoption in the auditing of higher education in developing economies, a qualitative approach was deemed most suitable for capturing the nuanced and context-dependent perspectives of audit practitioners (Poth, 2018). The interpretivist lens enables the researcher to explore how organizational culture, technological readiness, and institutional structures shape the experiences and perceptions of internal auditors (Ragubeer, 2022). This design provided flexibility to probe deeply into participants' views, in order to reveal the socio-technical and regulatory complexities influencing the integration of AI in the public university context.

3.1 Population and Sampling

The target population in this study comprised internal auditors, audit directors, and senior audit officers in Ghana's public universities, which are mandated to ensure accountability, transparency, and financial prudence in

line with the Public Financial Management Act (Act 921) and the Internal Audit Agency Act (Act 658). Public universities were selected because they manage large and complex financial systems, and they are increasingly under pressure to adopt digital technologies for effective governance. A purposive sampling technique was employed to select respondents with extensive knowledge and practical experience in internal auditing, digital transformation, and the use of emerging technologies such as AI.

Six universities were purposively chosen to ensure diversity in institutional size, digital maturity, and geographic distribution: the University of Ghana (UC), Kwame Nkrumah University of Science and Technology (KNUST), University of Cape Coast (UCC), University of Education (UE), Winneba, Ghana Communication Technology University (GCTU), and the University of Energy and Natural Resources (UENR). From each university, one key informant, either the Head of Internal Audit or a senior audit officer, was selected based on their professional experience, role in decision-making, and exposure to ICT and audit digitalization initiatives. To enhance the adequacy of the sample, the study captured interviewees with diverse functional roles and substantial professional experience. Participants ranged from mid-level auditors with 8 to 12 years of practice to audit directors with over 15 to 20 years of experience. Their responsibilities covered financial audits, risk-based audits, ICT-related audit tasks, compliance assessment, and advisory functions. Selection was further guided by professional credentials, such as membership in the Institute of Chartered Accountants in Ghana, the Association of Chartered Certified Accountants, or the Institute of Internal Auditors. Participants were contacted through official email invitations, followed by confirmation of consent and interview scheduling. Data saturation was achieved after the sixth interview. Consistent with Guest et al. (2020), saturation in homogeneous expert samples typically occurs within the first 6 to 12 interviews; in this study, no new codes emerged after the fifth interview, and the sixth interview produced full repetition of existing codes and concepts. This clear redundancy in thematic patterns provided strong evidence that additional interviews would not yield new insights, thereby justifying the final sample size. This sampling strategy ensured that the perspectives obtained were credible, contextually grounded, and reflective of internal auditing practices across major public universities in Ghana.

3.2 Collection of Data

Data were collected through semi-structured interviews designed to generate rich and detailed insights into how internal auditors perceive and experience the integration of AI within public university audit functions. This approach enabled participants to express their views freely while giving the researcher the flexibility to probe emerging issues for clarity and depth (Anim et al., 2025; Poth, 2018). The interview guide was developed directly from the constructs of the TOE framework and DOI theory to ensure theoretical consistency. Questions relating to system compatibility, infrastructure adequacy, and technological complexity were derived from the technological dimension of the TOE framework, while issues concerning leadership support, organizational culture, availability of resources, and audit autonomy were informed by its organizational dimension. Pressures from external policy, expectations from regulations, and environmental constraints shaped questions grounded in the TOE environmental context. The DOI theory informed questions on perceived relative advantage, compatibility with existing audit practices, trialability, observability of benefits, and perceived ease or difficulty of using AI tools. The interview guide covered 6 domains including technological readiness (infrastructure, compatibility, and complexity); organizational culture and leadership support (innovation climate, decision-making, and resources); auditors' competencies (skills and training); environmental and regulatory pressures (policies and compliance); DOI constructs (relative advantage, compatibility, trialability, observability, and ease of use); and ethical or data governance concerns (privacy, security, and accountability).

Ethical approval was secured before data collection, and formal authorization was granted by each participating university. Participants were provided with information documents outlining the objectives of the study, confidentiality measures, and their right to withdraw at any time. Interviews were carried out from May to July 2025, either in person or via Zoom and Microsoft Teams, according to participants' preferences. Each interview, lasting for approximately 45 to 60 minutes, was conducted with prior consent and audio-recorded. It was complemented by field notes capturing contextual observations. To augment credibility and triangulate findings, pertinent institutional documents, including internal audit reports, ICT policies, and digitalization strategies, were also examined. All recordings were transcribed verbatim and anonymized to ensure the protection of participants' identities.

3.3 Data Analysis

The data were analyzed by thematic analysis, which offered a structured yet flexible approach for identifying and interpreting patterns of qualitative data (Braun & Clarke, 2019). The analysis comprised six iterative stages, including familiarization through repeated review of transcripts, development of initial codes, identification of potential themes, evaluation and refinement of these themes, definition and naming of each theme, and ultimately, synthesis into a cohesive analytical narrative. NVivo 14 software enabled efficient data organization, precise

coding, and enhanced transparency throughout the analytical process. The coding and theme-generation procedures were guided by the TOE and DOI frameworks to ensure theoretical coherence. Deductive codes informed by TOE were created for technological readiness, organizational capacity, and environmental pressures, while DOI-based codes captured perceptual and behavioral elements such as perceived relative advantage, complexity, compatibility, and trialability of AI tools. During the development of themes, related codes were clustered into higher-order categories to be aligned with these frameworks. The emergent themes reflected both participants' perspectives and established theoretical constructs. This hybrid deductive-inductive strategy enabled the analysis to remain grounded in the lived experiences of auditors, while maintaining strong conceptual alignment. To ensure trustworthiness, member-checking was performed to allow participants to review and validate key interpretations (Shuwaili et al., 2024). Triangulation was achieved by comparing interview data with institutional documents and policy materials, strengthening dependability, and confirming the consistency of emergent themes.

4. Results

Thematic analysis of the interview data revealed four dominant themes explaining the challenges internal auditors face in integrating AI into internal audit functions in the public universities of Ghana. These themes include technological readiness and infrastructure constraints, organizational culture and leadership inertia, capacity and competency gaps, and regulatory and ethical ambiguities derived from iterative coding and interpretation using NVivo 14. Each theme reflects the complex interplay of technological, organizational, and environmental factors as conceptualized by the TOE and DOI frameworks. The diagram below displays the thematic map from the analysis.

Figure 1 was developed through a systematic qualitative mapping process based on the thematic analysis of interview transcripts. The researcher first coded all transcripts in NVivo, using both deductive codes derived from the TOE framework and the DOI theory, together with inductive codes emerging directly from participants' narratives. These codes were then grouped into four higher-order categories that consistently appeared across the cases: technological readiness and infrastructure constraints, organizational culture and leadership inertia, capacity and competency gaps, and regulatory and ethical ambiguities. The figure summarizes how these themes interact to shape AI integration within internal audit functions. The arrows represent relationships observed in the data, illustrating how contextual conditions either facilitate or constrain adoption. Thus, the figure serves as a conceptual synthesis of empirical evidence and theoretical interpretation.

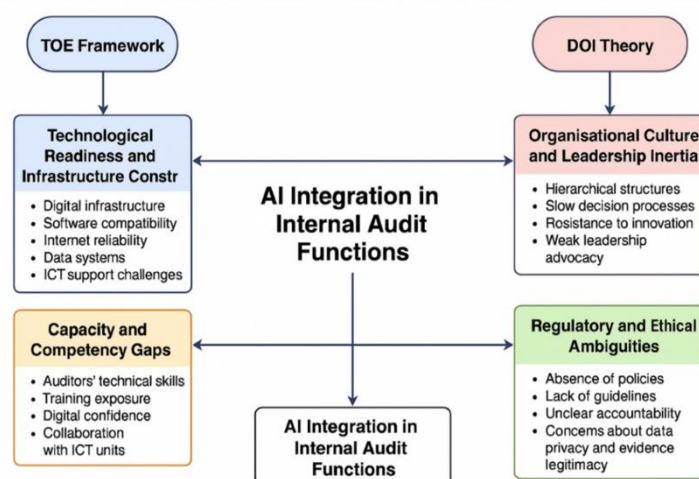


Figure 1. Themes related to the challenges faced by internal auditors when integrating AI

4.1 Technological Readiness and Infrastructure Constraints

Participants consistently reported that inadequate technological infrastructure was one of the most significant barriers to integrating AI into internal audit functions. They described outdated systems, unreliable networks, fragmented data storage, and the absence of AI-compatible tools as major obstacles. One auditor noted: "Our systems are not connected. Everything is scattered, and you cannot run any advanced tool on what we have." (UG)

This limited digital environment meant auditors relied heavily on manual processes, which restricted efficiency and reduced confidence in the feasibility of adopting AI. Several respondents also highlighted the challenge of securing institutional investment for upgraded audit technologies. Procurement processes were described as

lengthy and unpredictable, thus delaying even basic ICT improvements. As one auditor from the UENR stated: “When you request new software or hardware, it can take years. By the time it arrives, it is already behind the times.”

These constraints created a sense of technological stagnation within audit units. Auditors additionally noted that the lack of dependable data systems rendered the incorporation of AI impractical under the present circumstances. Without secure storage solutions, integrated databases, or immediate access to digital records, participants regarded AI-enabled auditing as technically unfeasible. One auditor from the UCC explained: “AI depends on data, but our data is either incomplete or not digital. So how can AI work here?”

Overall, these accounts reflect an environment where the limitations of infrastructure severely restrict readiness for the implementation of AI.

4.2 Capacity and Competency Gaps

All participants emphasized substantial capacity constraints as a primary obstacle to the adoption of AI. Respondents often highlighted the disparity between their academic preparation and the demands of contemporary and technology-driven auditing practices. Numerous internal auditors have been trained primarily in conventional accounting frameworks with limited exposure to digital analytics. One auditor from the GCTU captured this sentiment clearly, stating: “We were trained for traditional auditing. AI requires skills most of us were never taught. When you mention coding or algorithms, people switch off because it feels like a different profession entirely.”

This gap left staff feeling unprepared, anxious, and sometimes overwhelmed. The insufficiency of institutional training further exacerbated these concerns. Participants characterized the ICT seminars as elementary, monotonous, and lacking practical applicability to their auditing responsibilities. As one auditor from the UCC clarified: “The ICT training we get focuses on basic computer use. Nothing targets AI or digital auditing. You leave the training the same way you came in, with new tools, no new skills to apply in our actual work.”

This absence of structured upskilling rendered auditors reliant on obsolete knowledge and informal learning methods. The psychological aspect of competency deficiencies was also notably significant. Many auditors were concerned that AI could undermine their significance or supplant key audit responsibilities. This concern was explicitly articulated by a participant from the UENR who stated: “There is fear that AI will replace us. People worry it might take over most of the audit work, especially the routine parts.”

This apprehension influenced perceptions of technology, resulting in staff hesitating to engage in digital transformation initiatives. Challenges of coordination exacerbated the situation further. Respondents observed that audit units and ICT departments infrequently exchanged knowledge and seldom collaborated, resulting in a siloed environment. One auditor from the UG noted: “We work in silos. If we had stronger collaboration with IT, maybe we could learn gradually, but right now everyone stays in their corner.”

This isolation impeded both technical education and organizational preparedness for AI.

4.3 Regulatory and Ethical Ambiguities

Participants expressed strong uncertainty regarding how AI-generated audit output fits within existing regulatory frameworks. Many auditors were unclear whether the Ghana Audit Service, the Public Financial Management Act, or institutional audit committees recognized AI tools as valid sources of evidence. One auditor from the KNUST explained: “There is no policy telling us if AI output is legitimate. So, everyone is afraid to rely on it. You do not want to produce an audit report based on something that has no official acceptance.”

This regulatory gap created hesitation, with many auditors opting to avoid AI rather than risk noncompliance. Ethical considerations further intensified this reluctance. Respondents expressed concerns regarding data security, confidentiality, and the potential for misuse or manipulation of information. One auditor from the UCC highlighted the fragility of current systems: “Our data systems are not secure enough. Bringing in AI might expose sensitive information, especially when our servers are already vulnerable.”

These concerns were amplified by the absence of transparent accountability frameworks, leaving auditors uncertain as to who would be held responsible if an AI system generated inaccurate or biased results. Participants also perceived limitations due to the lack of national guidance or institutional governance frameworks supervising AI in public auditing. As an auditor from the UEW noted: “Until the government offers definitive guidance or financial support, AI will continue to be a concept rather than an established practice. We are unable to advance initiatives lacking regulatory approval or financial assistance.”

Without alignment to national policies, universities continued to exercise caution, favoring conventional manual audit procedures, which were regarded as more secure and compliant. Besides, auditors expressed concerns about professional liability, especially in cases where AI-generated insights might conflict with human judgment. One auditor from the UG remarked: “We need ethical guidelines and standards before using AI. Otherwise, if something goes wrong, the blame falls entirely on the auditor.”

This sense of personal exposure made auditors wary of participating in pilot projects or digital experiments.

Overall, regulatory deficiencies and ethical ambiguities undermined institutional confidence, fostering a cautious environment where AI was regarded as hazardous, insufficiently regulated, and potentially detrimental to audit integrity.

4.4 Organizational Culture and Leadership Inertia

A dominant theme that emerged across all institutions was the restrictive and bureaucratic nature of organizational culture. Participants described environments characterized by rigid procedures, multiple administrative layers, and slow decision-making processes that stifled innovation. One auditor from the UCC described the frustration vividly: “Everything requires layers of approval. Innovation dies in the queue. By the time a simple idea goes through all the committees, people have lost interest or the technology has moved on.”

These procedural bottlenecks discouraged initiatives and reinforced a preference for maintaining routine practices. Leadership behavior also played a significant role in shaping perceptions of adopting AI. Participants overwhelmingly stated that senior management did not prioritize technological advancement; instead, they focused on compliance and familiar workflows. As one auditor from the UG explained: “Management is comfortable with the old system. If leaders do not push for innovation, nothing moves. People follow the tone from the top, and the tone is usually ‘stick to what we know’.”

This leadership posture created a ripple effect, dampening enthusiasm and signaling that digital innovation was not an institutional priority. Another challenge raised was the absence of internal champions, individuals with influence, and digital insight into who could promote and sustain technology-driven change. One respondent from the UENR noted: “No one is leading digital innovation. Without a champion, people stick to what they know because no one is showing the way.”

The lack of mentorship, advocacy, and strategic communication around technology reinforced resistance among staff. Some auditors further explained that morale remained low because staff felt unsupported and under-resourced. Another respondent from the KNUST commented: “We need leaders who understand technology and can motivate us to embrace it. Otherwise, people will keep doing what they have always done.”

Without strategic leadership, initiatives of staff development stalled; innovative ideas remained unexecuted, and digital transformation became fragmented or symbolic rather than substantive.

4.5 Discussion

The results of this study indicate that technological readiness continues to be the most significant factor in the implementation of AI within internal audit functions at public universities in Ghana. The results indicate that the majority of institutions do not possess the essential infrastructure necessary to facilitate AI-enabled auditing, such as secure cloud storage, reliable data systems, and integrated analytics software. Participants consistently reported that internal audit units utilized manual or semi-digital tools, hence storing records in physical files or Excel, which restricts data accessibility and efficiency. This technological underdevelopment is consistent with the technological dimension of the TOE framework, which proposes that the availability, complexity, and compatibility of technology determine the readiness for adopting AI (Adorm-Takyi, 2023). These findings are consistent with those of Celestin & Gidisu (2023a) and Smith et al. (2025), who discovered that digital capacity deficits continued to impede technological advancement in public institutions. Furthermore, weak internet connectivity, outdated ICT infrastructure, and inadequate IT support were identified as obstacles. Auditors’ perception of AI as complex and incompatible with existing systems contributes to low adoption intentions from the DOI perspective (Auwah et al., 2022). Therefore, the deployment of AI in auditing will be fragmented and experimental rather than transformative in the absence of a comprehensive digital infrastructure and institutional prioritization.

The study also discovered that the integration of AI into internal audit processes was significantly influenced by organizational culture and leadership attitudes. Participants characterized the hierarchical and bureaucratic structure of Ghana’s public universities as resistant to change, as they frequently prioritize compliance and cost-control over innovation. This observation is consistent with Afadzinu et al. (2024), who contended that the TOE framework was fundamentally comprised of managerial commitment and resource allocation. Digital experimentation is discouraged, and inventive problem-solving is stifled by leadership inertia, resulting in audit departments being under-supported. The diffusion of new technologies is impeded by the absence of influential champions for change from the DOI perspective, as individuals emulate the behavior and priorities of their leaders (Obuobi, 2024). Similarly, Ataribanam (2024) emphasized that the perception of AI as either a threat or an enabler was contingent upon the support of senior management. Participants observed that the approval processes for innovation proposals are frequently protracted, which serves as an illustration of how structural rigidity impedes technological advancement. As a result, the cultivation of an innovation-driven culture necessitates a sustained managerial commitment to digital transformation, flexible administrative systems, and visible leadership advocacy.

Human capacity and competency gaps were also identified as central barriers to integrating AI. Respondents

emphasized that the majority of internal auditors lacked the technical literacy necessary to effectively collaborate with IT professionals, interpret data analytics, or operate AI tools. This deficiency is in accordance with the findings of Badu (2025) and Celestin & Gidisu (2023b), which indicate a discrepancy between the requirements of modern digital auditing and traditional accounting education. Participants characterized their training as generic and infrequent, with minimal emphasis on AI-driven audit simulations or applied data analysis. The TOE framework emphasizes the importance of human capital as a critical organizational determinant of adoption preparedness, whereas the DOI theory links skill scarcity to reduced trialability and increased perceived complexity (Sarpong, 2021; Tetteh et al., 2021). The subsequent apprehension regarding automation and job displacement serves to fortify opposition to change. To surmount this obstacle, auditors must possess hybrid competencies that combine technological proficiency with auditing expertise. AI-focused training and certification could be facilitated through collaboration with professional entities such as the Institute of Internal Auditors and the Institute of Chartered Accountants in Ghana. This would guarantee that audit teams are equipped to engage with emerging technologies confidently and ethically. Finally, the study found that regulatory uncertainty and ethical ambiguities constituted major external constraints on AI adoption.

Participants expressed apprehension regarding the absence of national or institutional policies that regulate the ethical use of AI in auditing, particularly in the areas of data privacy, algorithmic accountability, and the admissibility of AI-generated evidence. This is indicative of the environmental context of the TOE framework, which underscores the influence of institutional and policy pressures on the decisions of innovation (Anokye, 2022). Respondents expressed concern that the adoption of AI could expose auditors to legal or ethical risks in the absence of explicit guidance from oversight authorities such as the Internal Audit Agency or the Ministry of Finance. These findings support Agana et al. (2025) and Ataribanam (2024), who observed that weak regulatory ecosystems discouraged experimentation in AI use in the public sector. Moreover, the lack of policy alignment has resulted in minimal funding for AI initiatives, relegating them to isolated pilot projects rather than strategic reforms. As Tetteh et al. (2021) contended, sustainable digital transformation necessitated concurrent investments in capacity development and regulatory governance. Thus, the absence of explicit ethical and regulatory frameworks not only impedes institutional confidence but also undermines national endeavors to modernize public financial management through intelligent auditing systems.

4.6 Emergent Suppositions

Drawing on the findings and their interpretation through the TOE and DOI frameworks, several suppositions emerge to explain the dynamics influencing AI adoption within internal audit functions in public universities in Ghana. These suppositions extend existing theoretical perspectives by illustrating how interrelated technological, organizational, and environmental factors shape adoption readiness and digital transformation outcomes in the public sector.

4.6.1 Technological readiness and AI adoption

Technological readiness directly determines AI adoption. The automation of audit processes is restricted by the absence of digital infrastructure, including secure data storage, reliable internet, and compatible software systems, at numerous public universities. Whenever technology is fragmented or obsolete, auditors resort to manual methods that diminish data accessibility and efficiency. The TOE framework underscores the fact that technological compatibility and availability are critical factors in determining adoption readiness (Anim et al., 2025). Consequently, organizations that possess sophisticated digital systems and technical support are more adept at effectively integrating AI tools. Continuous investment in ICT infrastructure is therefore indispensable for the implementation of transformative auditing practices.

4.6.2 Organizational culture and leadership commitment

The results suggested that the adoption of AI was significantly influenced by organizational culture and leadership attitudes. In public universities, technological innovation is frequently impeded by hierarchical and bureaucratic systems, which prioritize compliance over creativity. Innovation-driven leaders inspire confidence and resource commitment, while leadership inertia discourages experimentation (Doe-Dartey & Valand, 2024). The diffusion of new technologies is accelerated by visible champions for change. AI tools are more likely to be adopted by institutions that prioritize collaboration, trust, and transparency among their personnel (Anim et al., 2025). Consequently, leadership commitment not only facilitates cultural transformation but also serves as a mediator between technological readiness and adoption outcomes:

S2: Organizational culture and leadership commitment mediate the link between technological readiness and AI adoption.

Figure 2 was developed through a systematic qualitative mapping process based on the themes that emerged from the analysis. First, all interview transcripts were coded in NVivo 14, using both deductive codes drawn from the TOE and DOI frameworks and inductive codes that emerged directly from participants' narratives. These codes

were then clustered into higher-order categories representing the four dominant thematic areas: technological readiness, organizational culture and leadership, human capacity and digital competence, and regulatory and ethical conditions. Relationships among the categories were identified by examining how participants described interactions between technological, organizational, and environmental factors. The conceptual framework was refined through iterative comparison between empirical patterns and theoretical propositions. Figure 2, therefore, represents a visual synthesis of the emergent suppositions (S1–S4), showing how each factor influences AI adoption within the internal audit functions of public universities in Ghana.

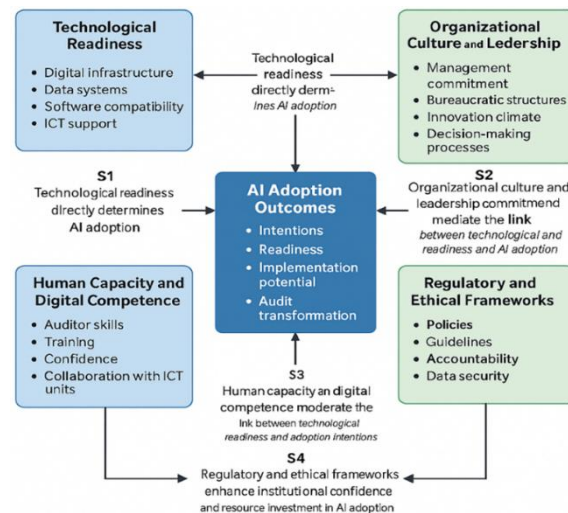


Figure 2. Proposed conceptual framework for the adoption of AI in internal audit functions of public universities in Ghana

4.6.3 Human capacity and digital competence

The study identified that a major obstacle to AI adoption was the limited digital literacy of internal auditors. Most auditors lack adequate training to interpret data analytics or operate AI systems confidently (Sheta et al., 2025). Without these skills, even advanced technology remains underused. The TOE framework highlights human capital as a key determinant of organizational preparedness, while the DOI theory links knowledge and trialability to successful adoption (Shuwaili et al., 2024). When auditors possess hybrid skills combining auditing and data analytics, they can better collaborate with IT professionals and trust AI-driven output. Ongoing professional development and AI-focused certifications are, therefore, essential for readiness.

S3: Human capacity and digital competence moderate the link between technological readiness and adoption intentions.

4.6.4 Regulatory and ethical frameworks

Participants expressed apprehension regarding the absence of explicit regulatory and ethical standards that regulate the use of AI in auditing. Auditors are concerned about the potential misuse of data, privacy violations, or accountability challenges in the absence of institutional or national policies. The environmental context of the TOE framework underscores the impact of external pressures, such as regulation, on decisions of innovation (Anomah et al., 2024). The establishment of appropriate ethical and policy frameworks fosters responsible experimentation, builds institutional trust, and attracts funding for AI initiatives. Auditors are more inclined to incorporate AI into their work when they perceive themselves as legally and ethically safeguarded.

S4: Regulatory and ethical frameworks enhance institutional confidence and resource investment in AI adoption.

4.7 Implications and Contributions

4.7.1 Theoretical contribution

This study extends the TOE framework and DOI theory to the underexplored context of internal auditing in public universities within a developing economy, thereby making a substantial theoretical contribution. Although previous research has utilized these models in private-sector or commercial domains (Adorm-Takyi, 2023; Tetteh et al., 2023), this study contributes to the theory by illustrating how technological readiness, organizational culture, leadership commitment, and regulatory context collectively influence the adoption of AI in public-sector audit functions. The study empirically supports the assertion of Kudo (2024) that technological complexity and compatibility influence adoption intentions, but it also incorporates leadership inertia and institutional bureaucracy as distinctive organizational inhibitors. Furthermore, it expands upon the DOI theory by emphasizing how

perceived risks and limited trialability exacerbate resistance to innovation in low-resource environments. The results also contribute to the emerging literature on AI governance (Afadzinu & Koltai, 2025; Issah et al., 2024) by contextualizing ethical and regulatory uncertainty within a framework of institutional adaptation. The study provides a multidimensional perspective on the impact of socio-technical and policy contexts on AI-driven transformation in public-sector auditing systems, thereby enhancing theoretical understanding by bridging the divide between innovation diffusion and institutional theory.

4.7.2 Contributions to knowledge

This study represents the first known empirical investigation of the challenges surrounding AI integration into internal audit functions across Ghana's public universities, to offer novel insights into how contextual factors hinder technological transformation in the public sector. Unlike previous studies that focused on corporate auditing or private organizations (Aikins et al., 2022; Oyedotun et al., 2025), this research uniquely examined AI adoption within an academic governance environment, where bureaucratic hierarchies, limited funding, and weak regulatory structures intersect. The uniqueness of this work is its integration of empirical field data with the TOE and DOI frameworks to develop a comprehensive conceptual understanding of the collective impact of digital readiness, leadership attitudes, and ethical considerations of innovation diffusion in resource-constrained institutions. The study offered a comprehensive understanding of the lived experiences of internal auditors who are navigating digital transformation by incorporating the voices of six prominent Ghanaian universities. In addition, it contributes to the body of knowledge regarding the preparedness of the public sector for AI in Africa, a region in which such research is still scarce (Anomah et al., 2024; Dawuda & Salman, 2024). The originality of this study is also derived from its multi-level analysis, which integrates technological, organizational, and environmental barriers into a single interpretive model. This enriches the global academic discourse on sustainable and equitable AI-driven auditing practices.

4.7.3 Policy and practical implications

The results of this study have a variety of practical and policy implications for the advancement of AI integration in internal audit functions within Ghana's public universities. First, policymakers should prioritize the digital transformation of audit systems by investing in secure data infrastructure, cloud technology, and AI-compatible software to improve transparency and efficiency. Secondly, universities should implement targeted training and capacity-building programs that provide internal auditors with the necessary competencies in digital ethics, data analytics, and AI literacy. Third, the leadership of public institutions should promote an innovation-oriented culture by streamlining bureaucratic procedures, promoting experimental projects, and providing consistent financial support for technological experimentation. National oversight authorities, including the Internal Audit Agency and the Ministry of Finance, should establish comprehensive AI governance frameworks that address ethical use, data protection, and algorithmic accountability from a regulatory perspective. Furthermore, the long-term sustainability and responsible innovation of AI systems could be guaranteed through collaboration among technology providers, professional associations, and academia. In addition to improving audit quality and institutional accountability, the implementation of these strategies would also ensure that Ghana's higher education governance is in accordance with international digitalization standards. Ultimately, the study emphasized the pressing necessity of cross-sectoral and coordinated policies that fortify institutional preparedness for future AI-driven auditing reforms and reconcile the digital divide.

5. Conclusions

This study concluded that the adoption of AI in internal audit functions within Ghana's public universities was shaped by four interrelated factors: technological readiness, organizational culture and leadership, human capacity, and regulatory clarity. The findings indicated that the lack of a secure digital infrastructure and integrated data systems constrained technological readiness. Therefore, the study determined that investing in advanced ICT infrastructure is crucial for facilitating AI-supported auditing. The findings additionally indicated that bureaucratic frameworks and passive leadership diminished institutional dedication to innovation. Consequently, the research determined that visible leadership endorsement and a culture that promotes experimentation are essential for ongoing digital transformation. Human resource deficiencies arose as a significant obstacle, with auditors lacking the necessary digital skills to operate or interpret AI tools. Therefore, the study concluded that targeted AI-focused training and collaboration with professional organizations are essential to enhance auditor readiness. Finally, regulatory and ethical uncertainties were identified as factors that erode institutional confidence in the adoption of AI. Therefore, explicit institutional and national governance policies are essential to delineate ethical boundaries, establish accountability standards, and specify permissible applications of AI-generated evidence. Overall, these findings highlight that the effective integration of AI necessitates coordinated enhancements in infrastructure, capacity, organizational culture, and policy.

5.1 Limitations

This study has several limitations that should be acknowledged. The utilization of a small and intentionally selected sample of six participants, while methodologically appropriate for qualitative research, restricts the range of perspectives gathered. The findings, therefore, indicate profundity rather than statistical generalizability. Furthermore, the research was conducted solely within Ghana's public universities, indicating that the findings may not comprehensively reflect the challenges of AI adoption in other public-sector organizations or universities in various nations. Organizational structures, regulatory frameworks, and technological ecosystems differ considerably across regions, and these contextual variations may impact the applicability of the findings. Furthermore, the study predominantly depended on self-reported experiences, which may be influenced by personal perceptions or institutional factors.

5.2 Suggestions for Future Research

Future research should expand this study by employing a comparative or longitudinal design to explore how AI adoption evolves across different public-sector institutions and regions in Ghana. Besides, scholars may investigate the influence of external stakeholders, including technology vendors and regulatory agencies, on the outcomes of its adoption. The proposed relationships within the TOE and DOI frameworks may be further validated through quantitative studies that employ structural equation modelling. In addition, to create context-specific models for responsible and sustainable AI integration, future research should investigate the ethical dimensions of AI auditing, specifically the issues of transparency, accountability, and bias.

Author Contributions

Conceptualization, J.C.W.; methodology, J.C.W.; formal analysis, J.C.W.; investigation, J.C.W.; data curation, J.C.W.; writing—original draft preparation, J.C.W.; writing—review and editing, J.C.W., M.D., and K.B.; visualization, J.C.W.; supervision, M.D. and K.B.; validation, M.D. and K.B. All authors have read and agreed to the published version of the manuscript.

Ethical Approval

Ethical clearance for the study was obtained from the University of KwaZulu-Natal ethics board under reference number HSSREC/00008064/2024. All participants provided informed consent before taking part, and confidentiality was strictly maintained. Interview recordings and transcripts were securely stored, with no identifiable information disclosed during analysis or reporting.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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